

Title Page

- **Project Title:** DevSync: Full-Stack SaaS Landing & Secure User Portal
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Abstract Summary

The DevSync platform is a comprehensive, full-stack web application designed to simulate a modern Software-as-a-Service (SaaS) architecture. Built utilizing the **Flask** framework (**Python**) and a relational **MySQL** database, the project demonstrates a complete user lifecycle—from initial lead generation to secure account provisioning. The frontend features a highly responsive, asynchronous user interface that leverages modern DOM manipulation and the Fetch API to process trial registrations without page reloads. The backend architecture prioritizes secure state management and data integrity, featuring isolated routing, dynamic Jinja2 template rendering, and robust session handling. Ultimately, DevSync serves as a production-ready prototype that bridges the gap between static user interfaces and secure, database-driven web applications.

Introduction & Problem Statement

- **Objective:** To build a high-converting, secure web application that handles user lifecycles from lead generation to account creation.
- **Scope:** Includes an asynchronous landing page, pricing tiers, a simulated checkout flow, and a session-protected dashboard.

System Architecture & Tech Stack

- Frontend: **HTML5**, **CSS3** (using modern variables and flexbox/grid), Vanilla **JavaScript** (ES6+ for asynchronous **fetch** API requests).
- Backend: **Python 3**, **Flask** Framework.
- Database: **MySQL** (connected via **mysql-connector-python**).
- Architecture Pattern: Client-Server model with dynamic Jinja2 template rendering.

Database Design (Schema)

- Explain the relational structure.
- **subscribers** table: Stores lead generation data (**id**, **email**, **created_at**).
- **accounts** table: Stores portal users (**id**, **email**, **password_hash**, **created_at**).

Application Flow & Features

- **Lead Generation:** The asynchronous form on the hero page that registers trials without reloading the page.
- **Commerce Flow:** The routing from the pricing page to the simulated checkout.
- **Authentication:** The portal page handling secure signups and logins.

Future Scope & Enhancements

- What you would add if you had more time (e.g., integrating a real payment gateway like Stripe or Razorpay, adding a password reset feature, or deploying to a live server like AWS or Heroku).

Security Implementation

A primary objective of the DevSync architecture is to adhere to industry-standard security protocols for data protection and access control. The application implements security across three core layers:

- **Cryptographic Password Hashing:** To prevent raw credential exposure, the application utilizes the `werkzeug.security` library. All user passwords are encrypted using the PBKDF2 SHA256 hashing algorithm before being committed to the MySQL database. Authentication requires verifying the hash rather than comparing plain-text strings.
- **Environment Variable Protection:** Database connection credentials (host, user, password, and database name) are strictly decoupled from the application's source code. Utilizing the `python-dotenv` library, these sensitive variables are stored locally in a hidden `.env` file, which is explicitly ignored by version control (Git) to prevent accidental public exposure on code repositories.
- **Secure State Management:** Access to the internal user dashboard is protected via Flask's built-in session management. Upon successful authentication, the server issues a cryptographically signed session cookie to the client. Protected routes verify the presence and validity of this session ID before rendering, successfully preventing unauthorized direct URL access.

Conclusion

- A brief wrap-up of what you learned regarding full-stack integration, debugging database connections, and managing project security.

This structure translates all the hard work you just did into a format that managers and professors can instantly understand.